



The M-CATs

SEMINAR 4

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Test today's
knowledge MCAT
style!



M-CATS

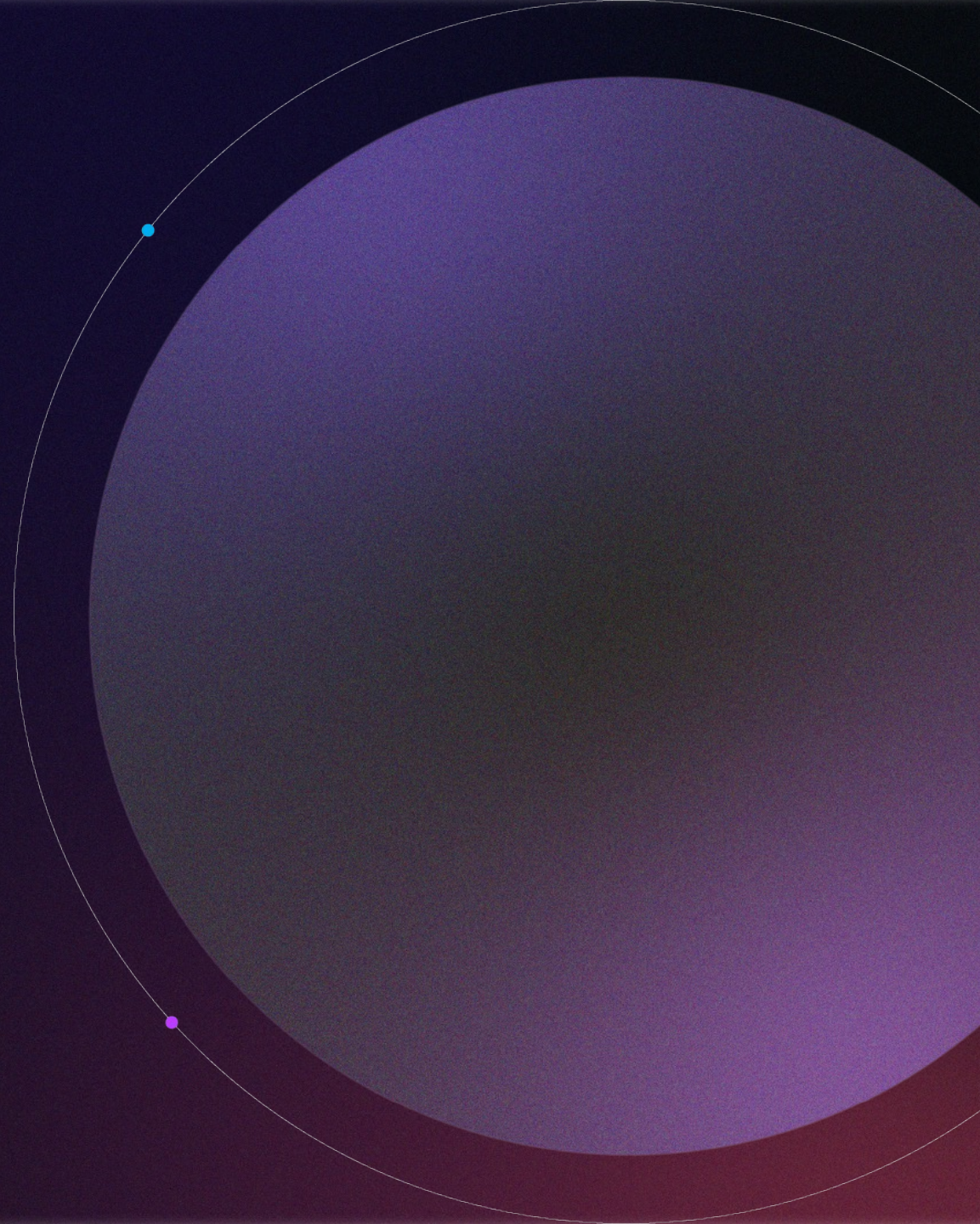
UNIVERSITY OF KENTUCKY

THE M-CATS PRESENTS

Isomers

ORGANIC CHAPTER 2

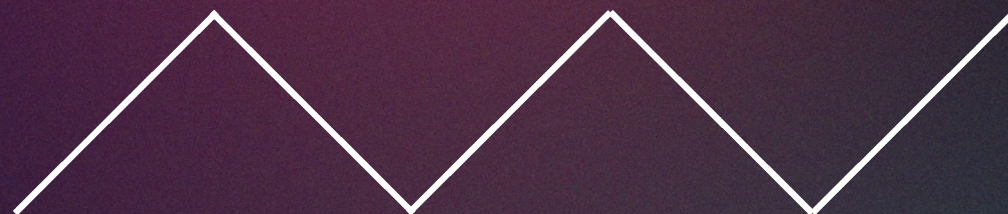
NIKHIL K



1 / 3

Da Types

Isomers



NOT ISOMERS

DIFFERENT MOLECULAR FORMULAS

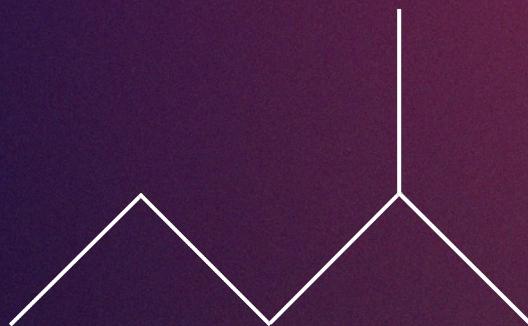
Isomers



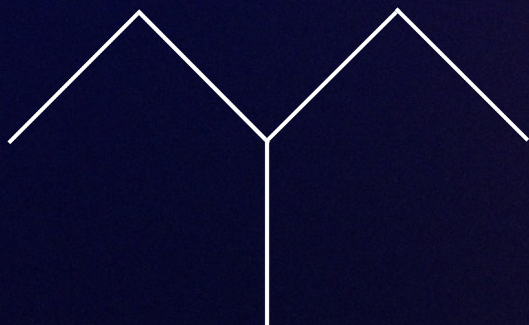
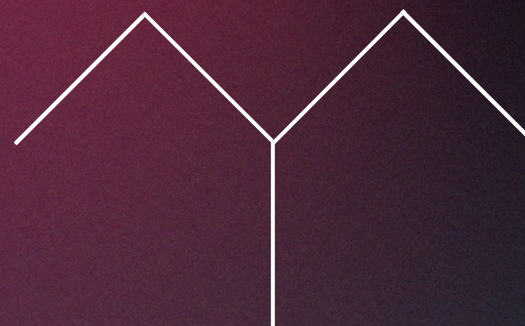
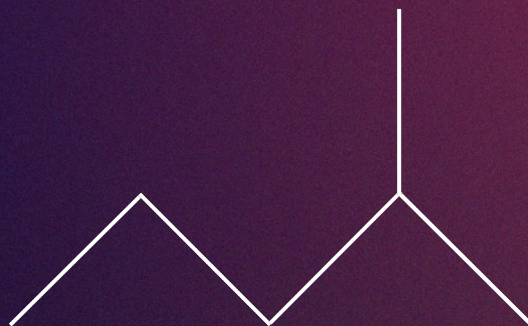
Isomers



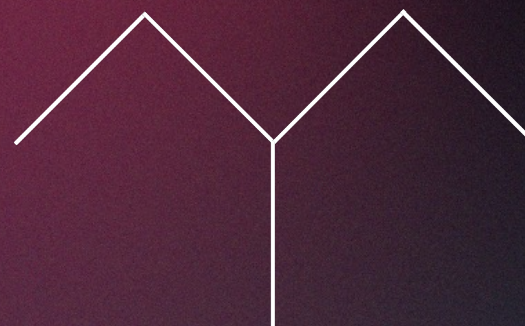
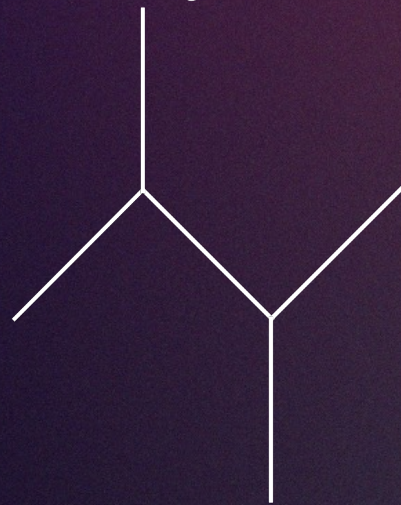
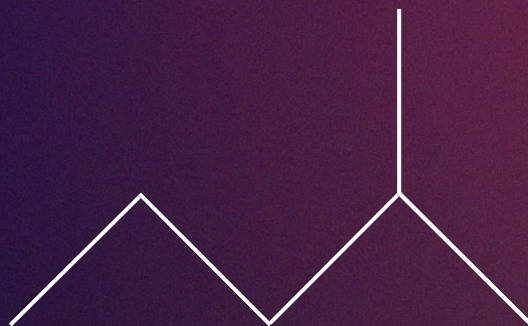
Isomers



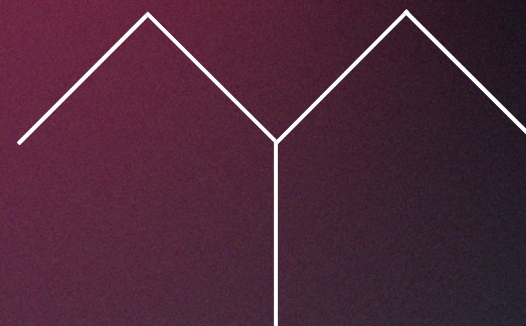
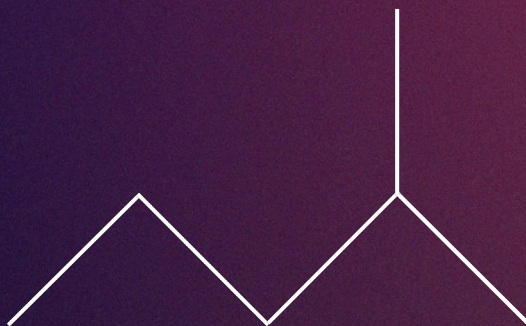
Isomers



Isomers



Isomers



Same Molecular
Formula,
Different
Structural
Arrangements

Isomers

Constitutional Isomers (Structural Isomers)

same molecular
formula but
**DIFFERENT
CONNECTIVITY**

Isomers

yes

?????

Same connectivity?

no

Constitutional Isomers
(Structural Isomers)

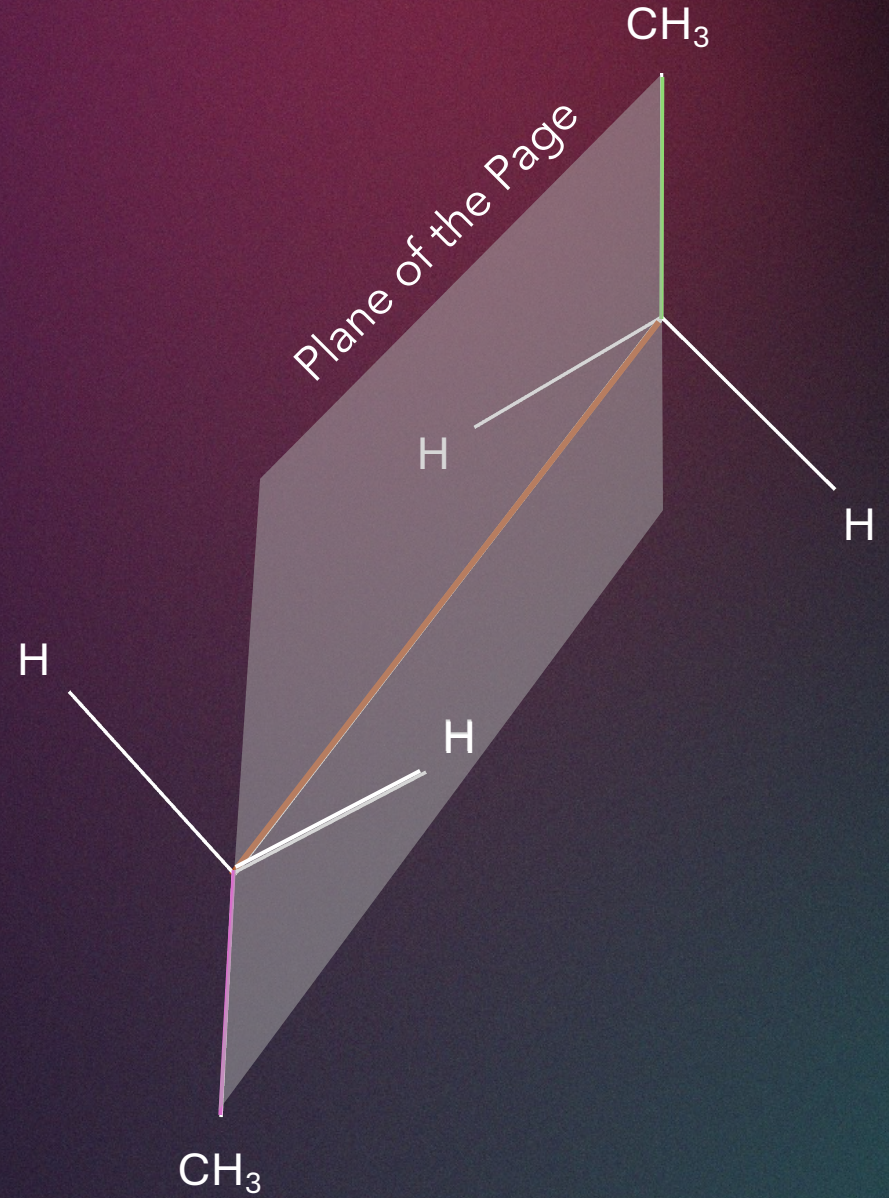
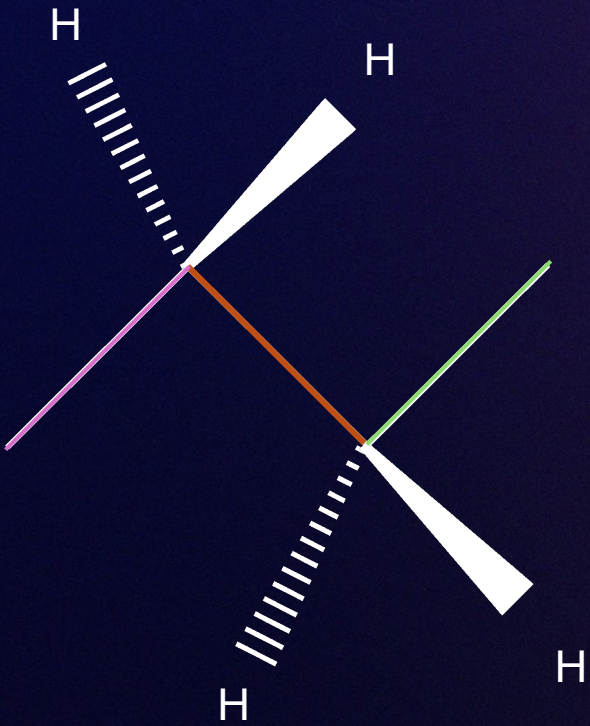
?????



Stereoisomer



Same molecular formula and connectivity but different arrangement in 3d space



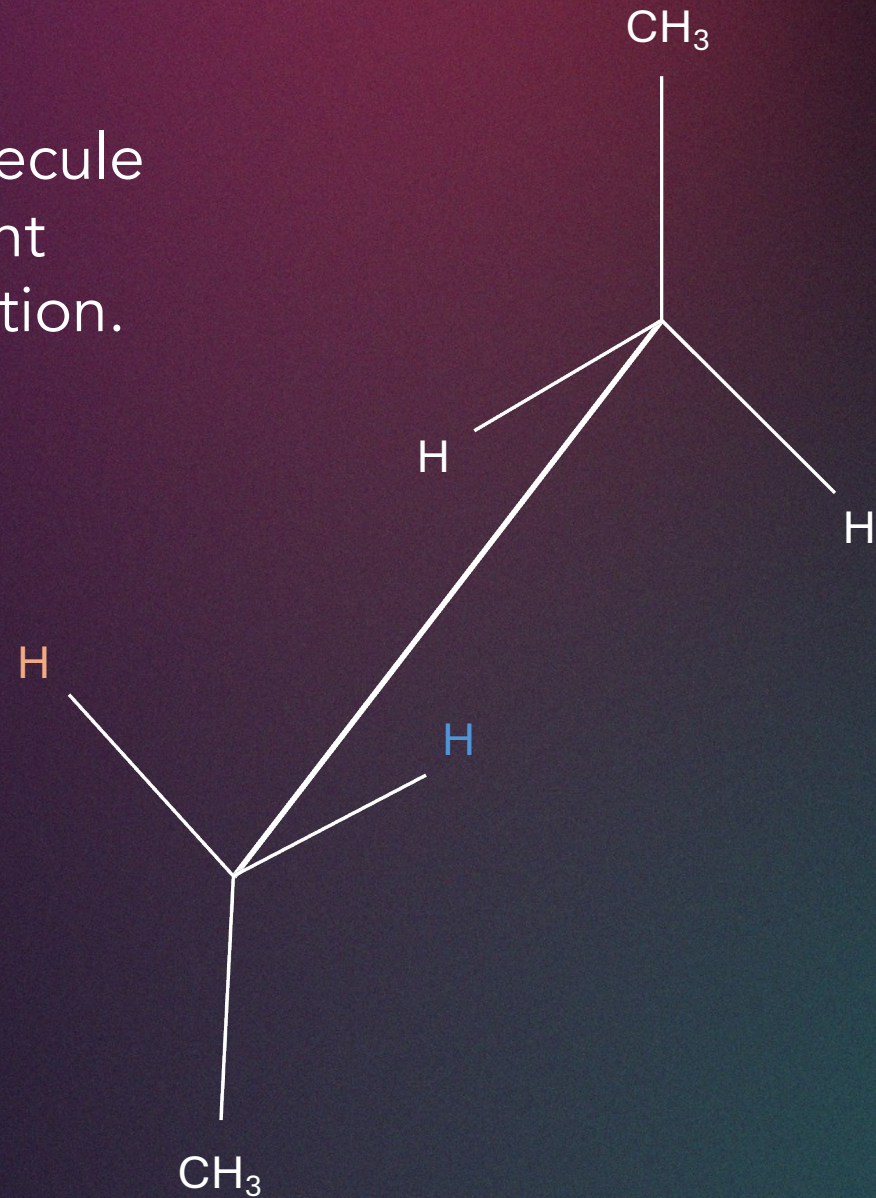
Stereoisomer



Same molecular formula and connectivity but different arrangement in 3d space



Same molecule
different
conformation.

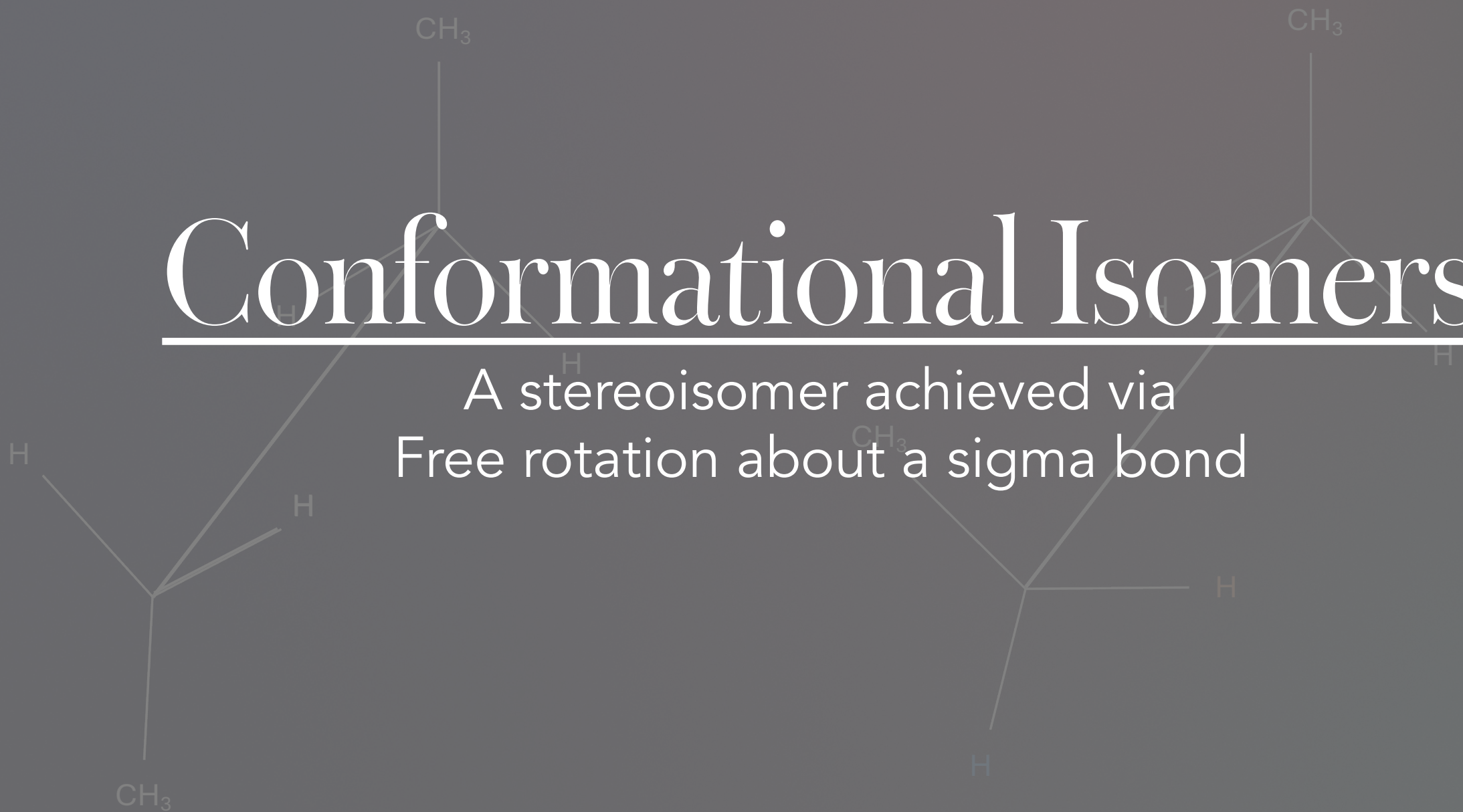


Stereoisomer

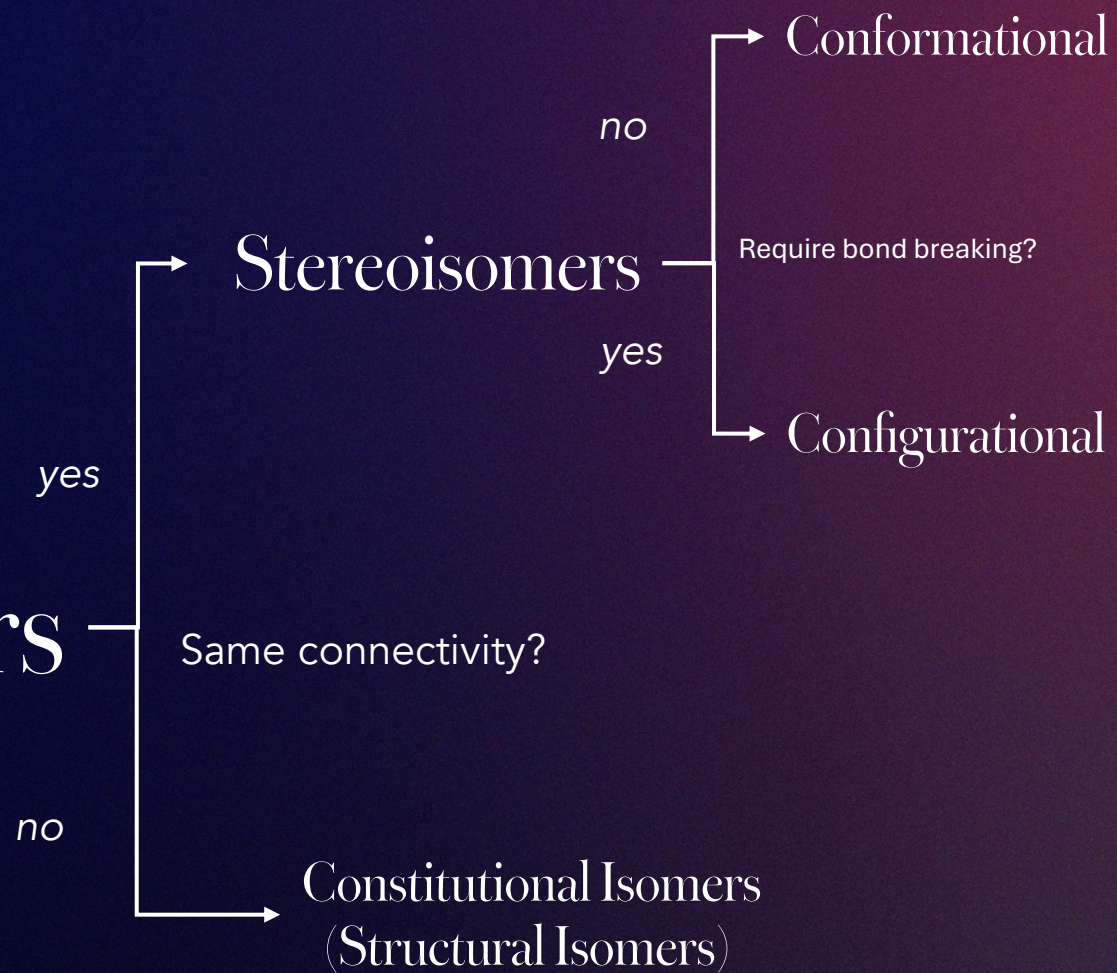
Same molecular formula and connectivity but
different arrangement in 3d space

Conformational Isomers

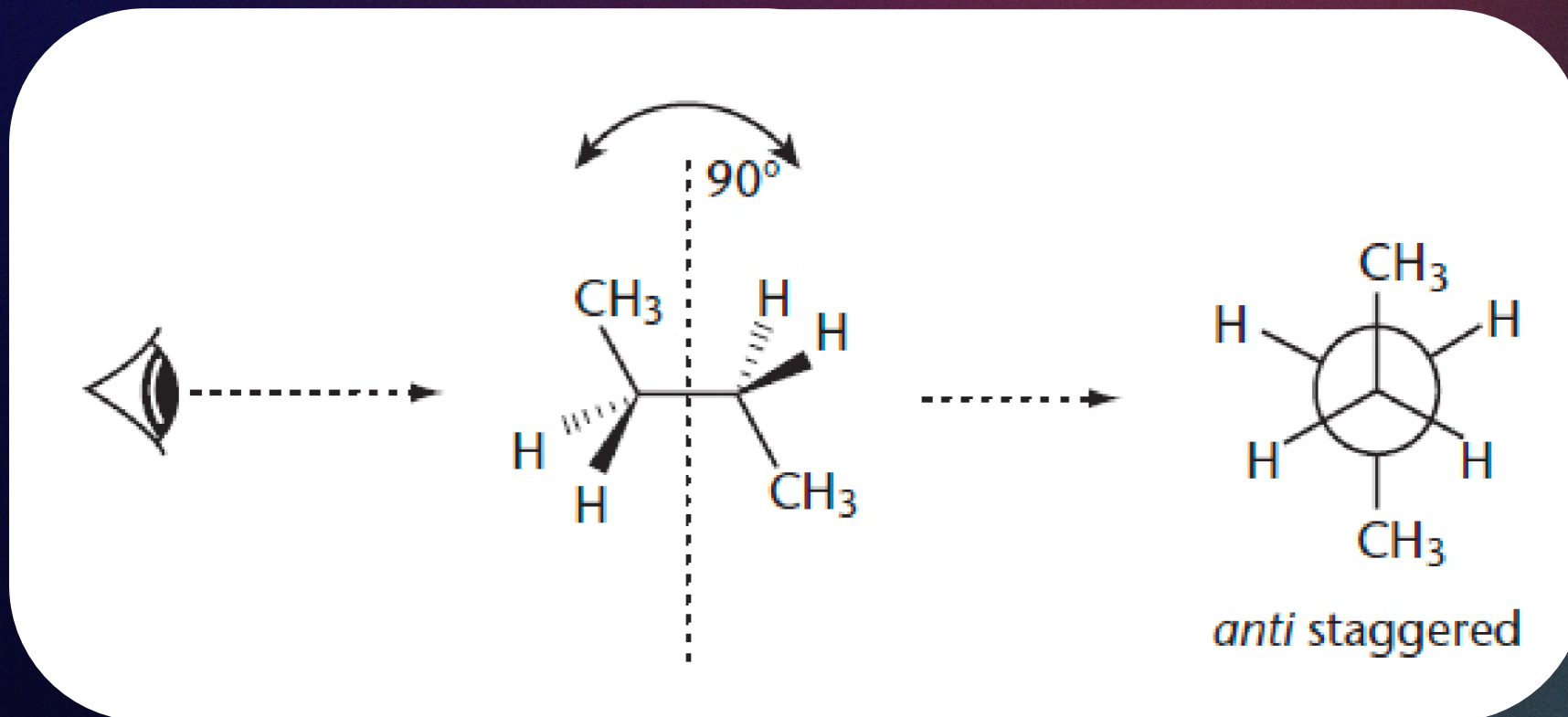
A stereoisomer achieved via
Free rotation about a sigma bond



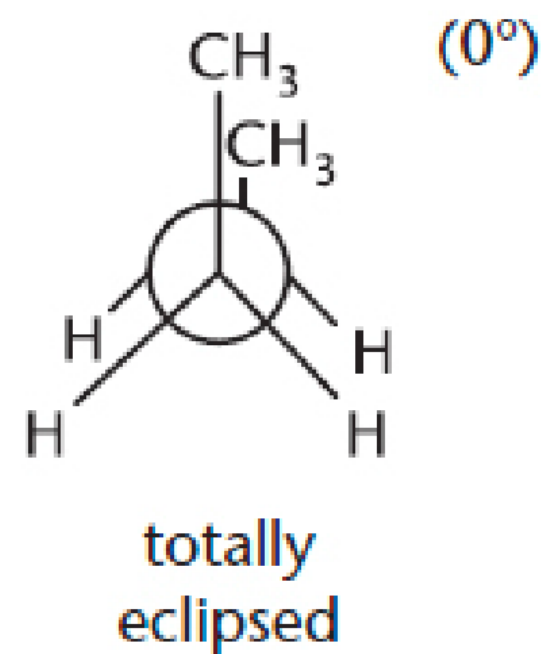
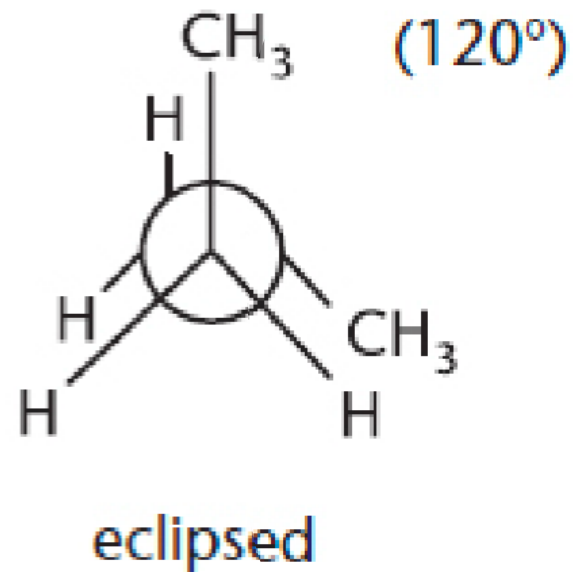
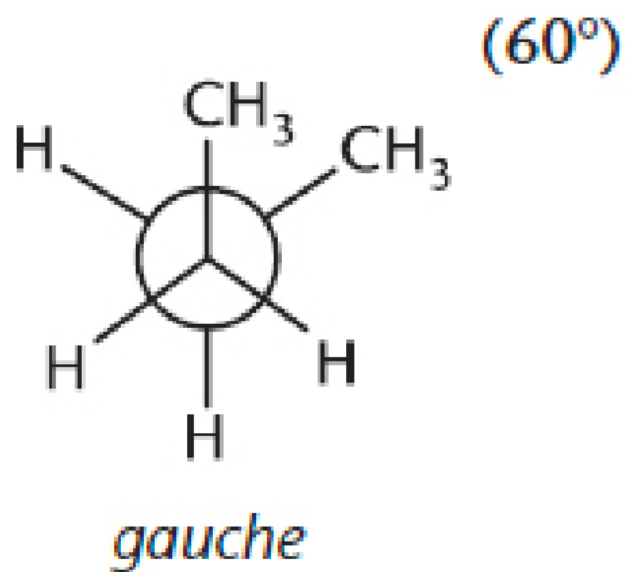
Isomers



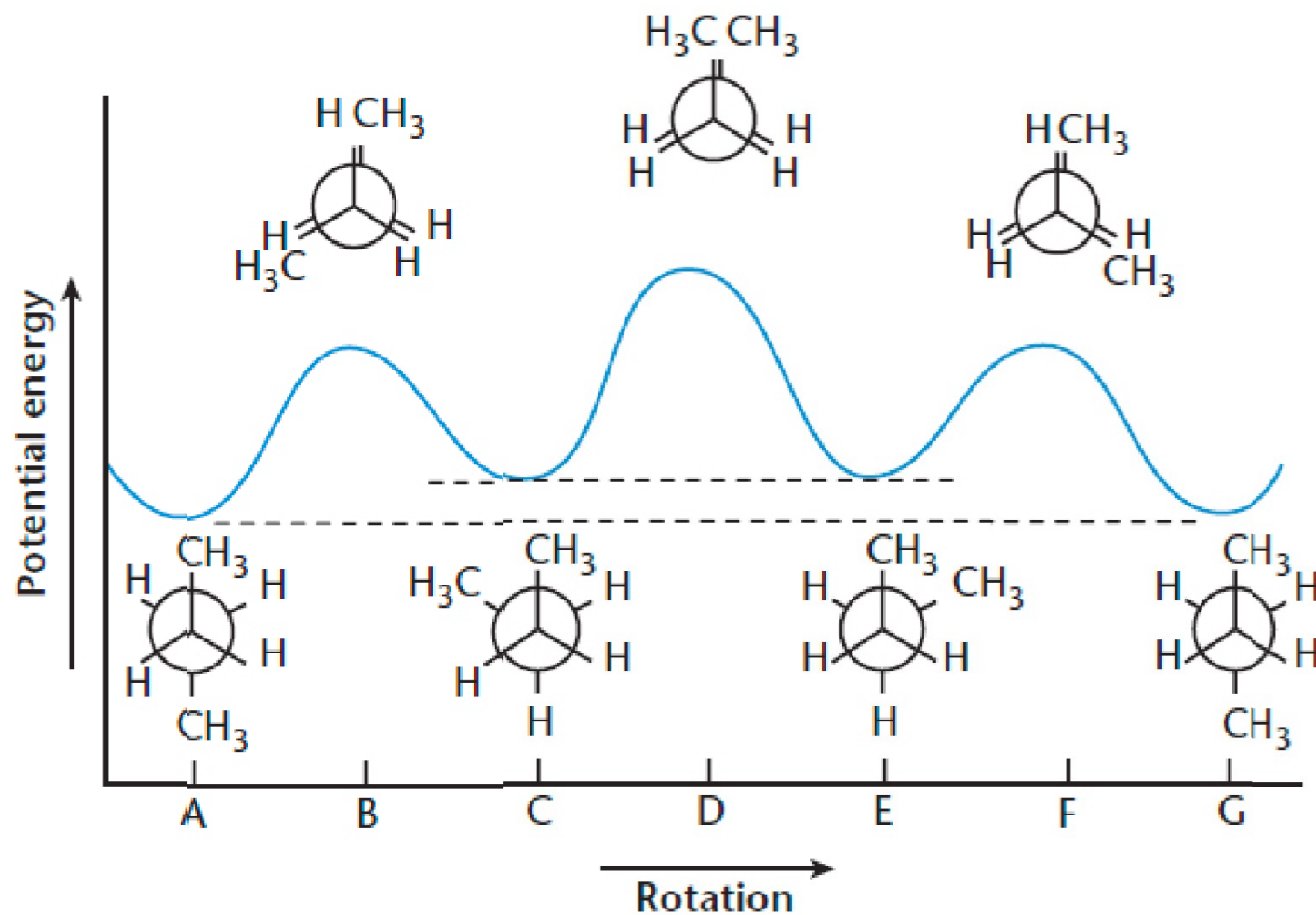
Conformational



Conformational

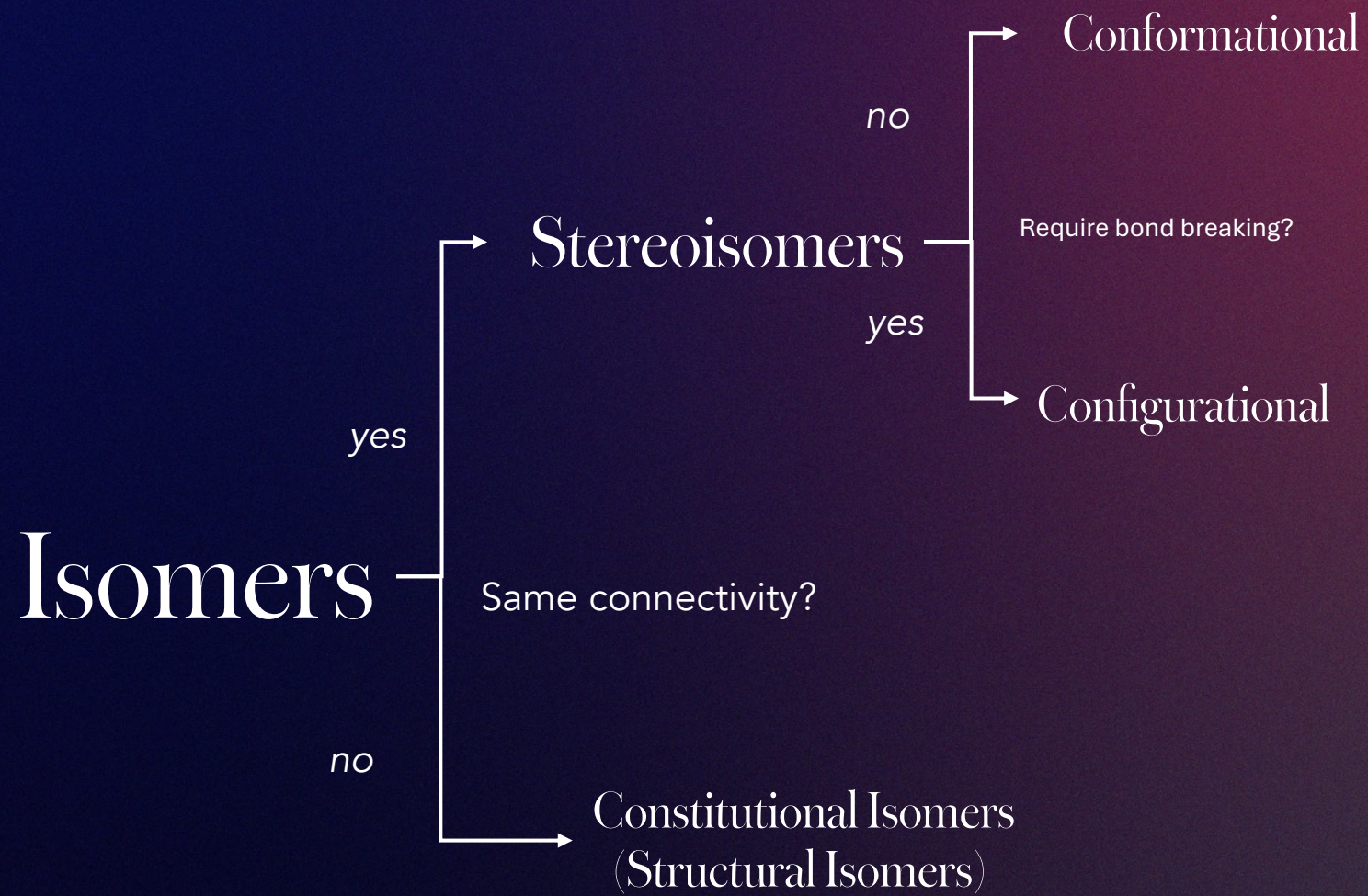


Conformational



Cyclic Conformations

Written Portion



Configurational

- Chirality
 - 4 unique groups
 - R versus S
 - Enantiomers & Diastereomers
 - Optical activity (dextrorotary and levorotary)
 - Specific Rotation equation
 - 2^n possible stereoisomers
 - Things you can infer regarding rotation w/ Diastereomers and Enantiomers

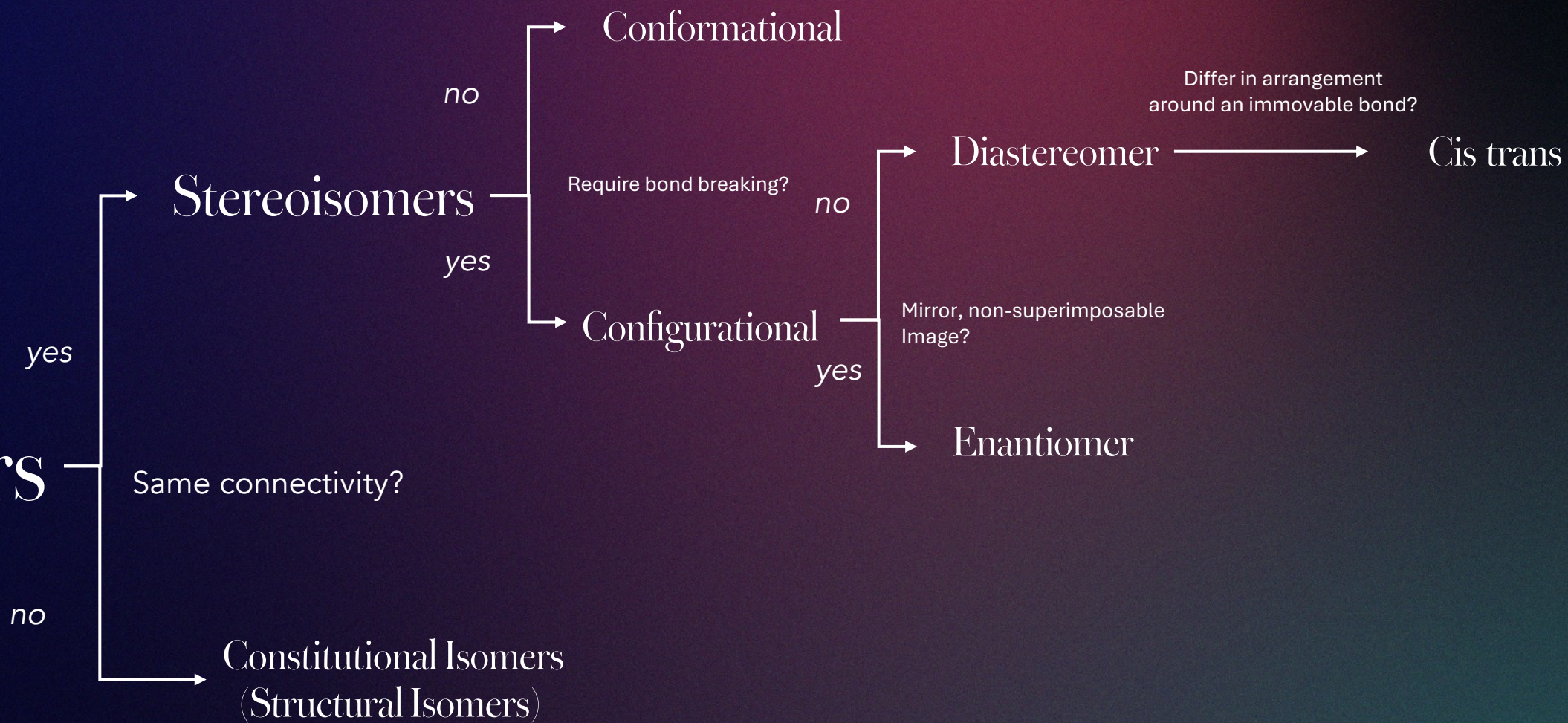
Configurational

- Chirality
 - 4 unique groups
 - R versus S
 - Enantiomers & Diastereomers
 - Optical activity (dextrorotary and levorotary)
 - Specific Rotation equation
 - 2^n possible stereoisomers
 - Things you can infer regarding rotation w/ Diastereomers and Enantiomers
 - Racemic compounds and meso compounds

Configurational

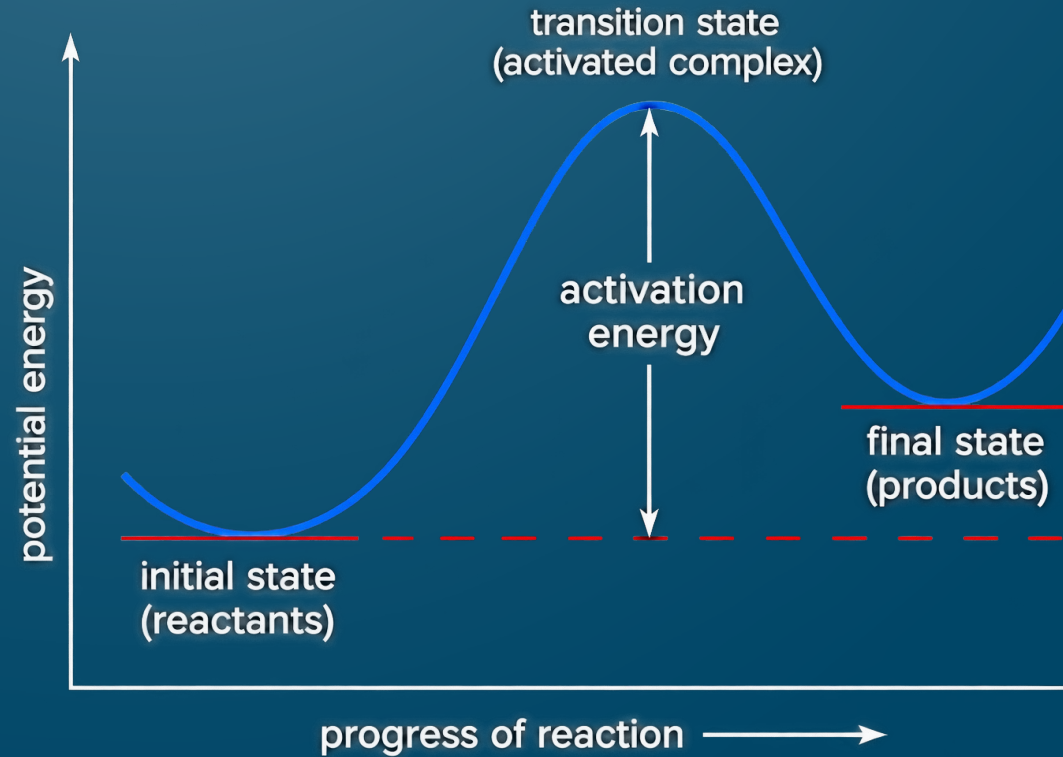
- R versus C
- Cis versus trans (double bond and cyclic alkane)
- E versus Z
- Fischer Projections

Isomers



General Chemistry Chapter V

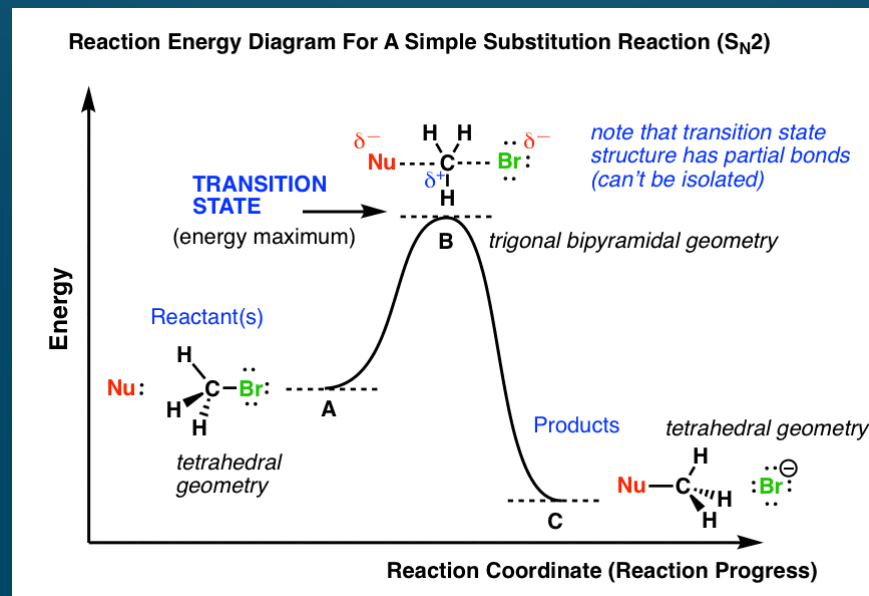
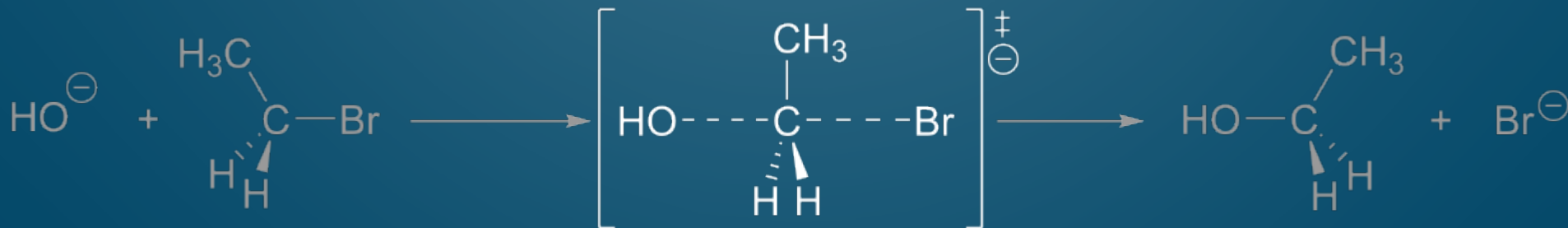
Chemical Kinetics



5.1: Chemical Kinetics



Transition State Theory



Collision Theory



$$\text{rate} = Z \cdot f$$

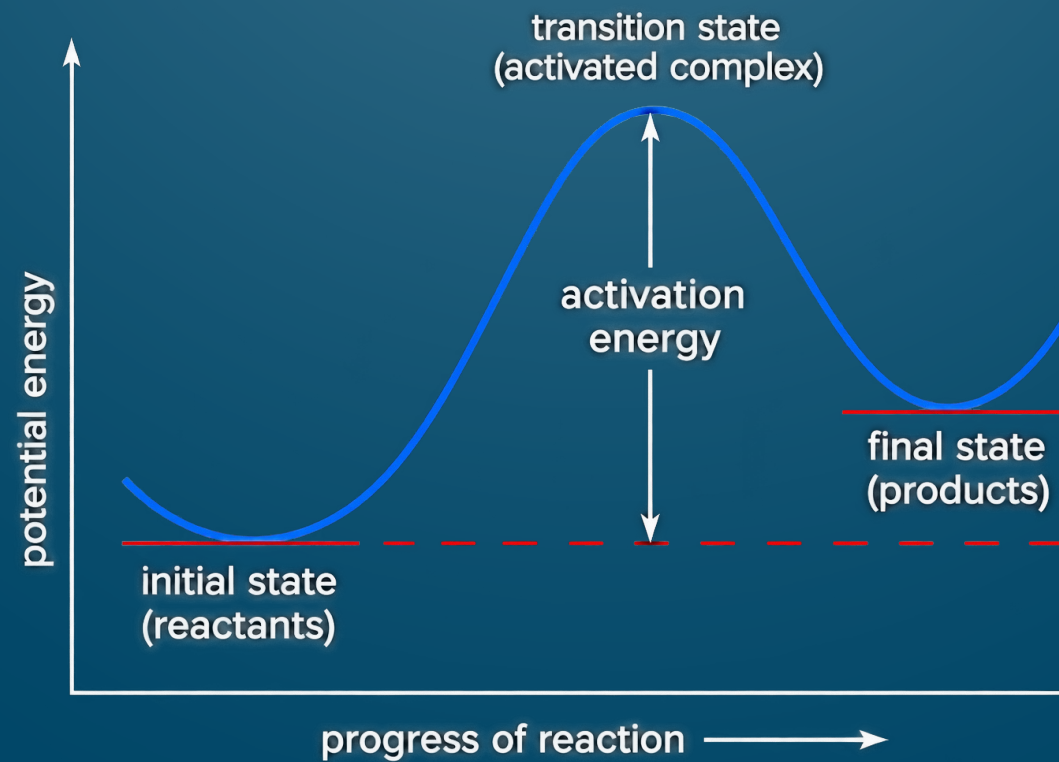
$$k = Ae^{-\frac{E_a}{RT}}$$

$$k \propto A$$

To increase k: $E_a \downarrow$, $T \uparrow$, $A \uparrow$



Free Energy of Reaction



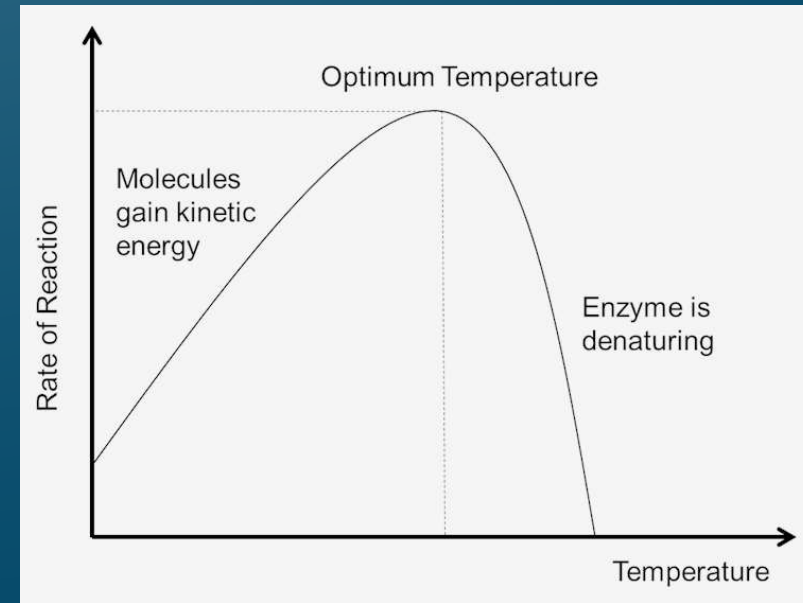
What Affects Reaction Rate

Concentration

Temperature...

Catalysts

Solvent



Reaction Rate

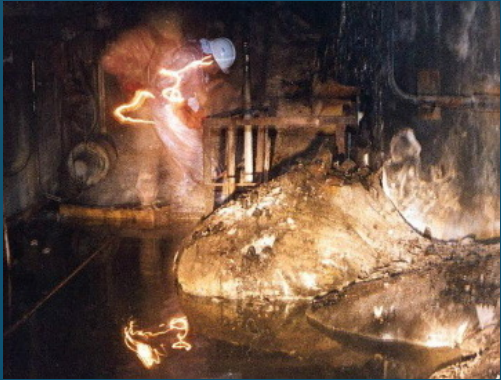
$$rate = \frac{\Delta product}{\Delta time}, -\frac{\Delta reactant}{\Delta time}$$



$$rate = k[A]^x[B]^y$$



5.2: Reaction Rates



Number of Sexual Partners	SEXUAL EXPOSURE CHART (If every person has only the same number of partners as you)	Number of People Exposed to
1		1
2		3
3		7
4		15
5		31
6		63
7		127
8		255
9		511
10		1023



Reaction Order



$$rate = k[A]^x[B]^y$$

$$order = x + y$$

X wrt A, y wrt B



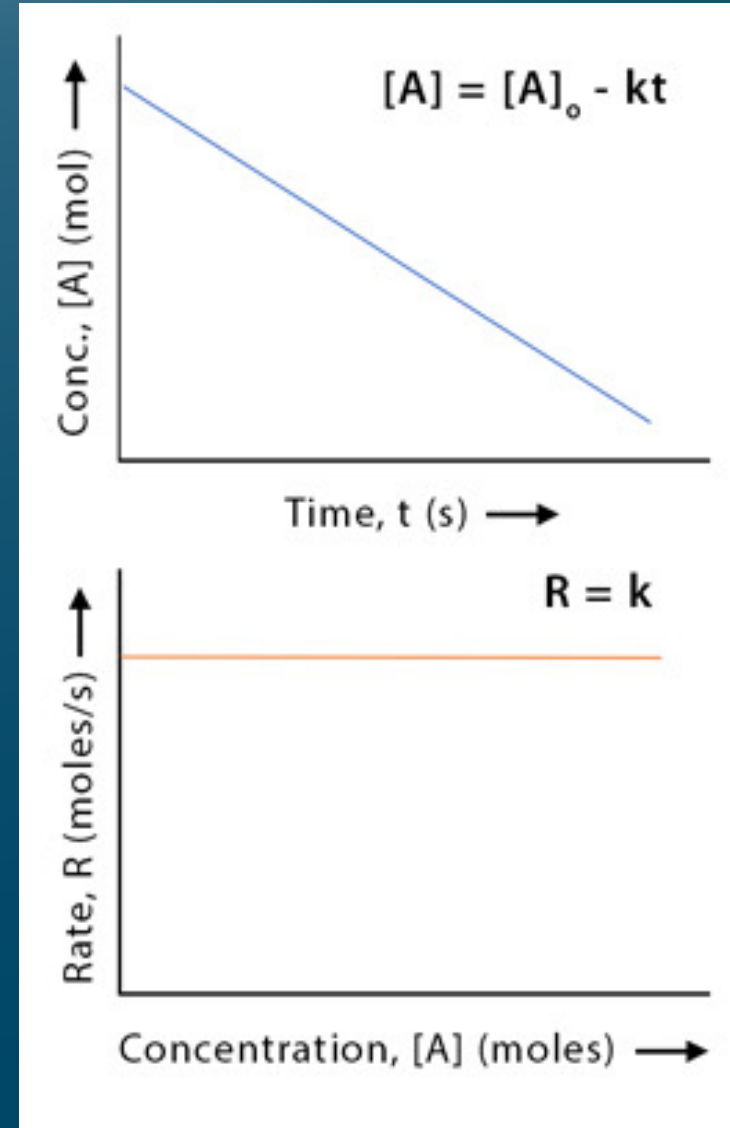
0th Order



$$rate = k[A]^0[B]^0$$

A and B rn

$$rate = k[A]^0[B]^0$$



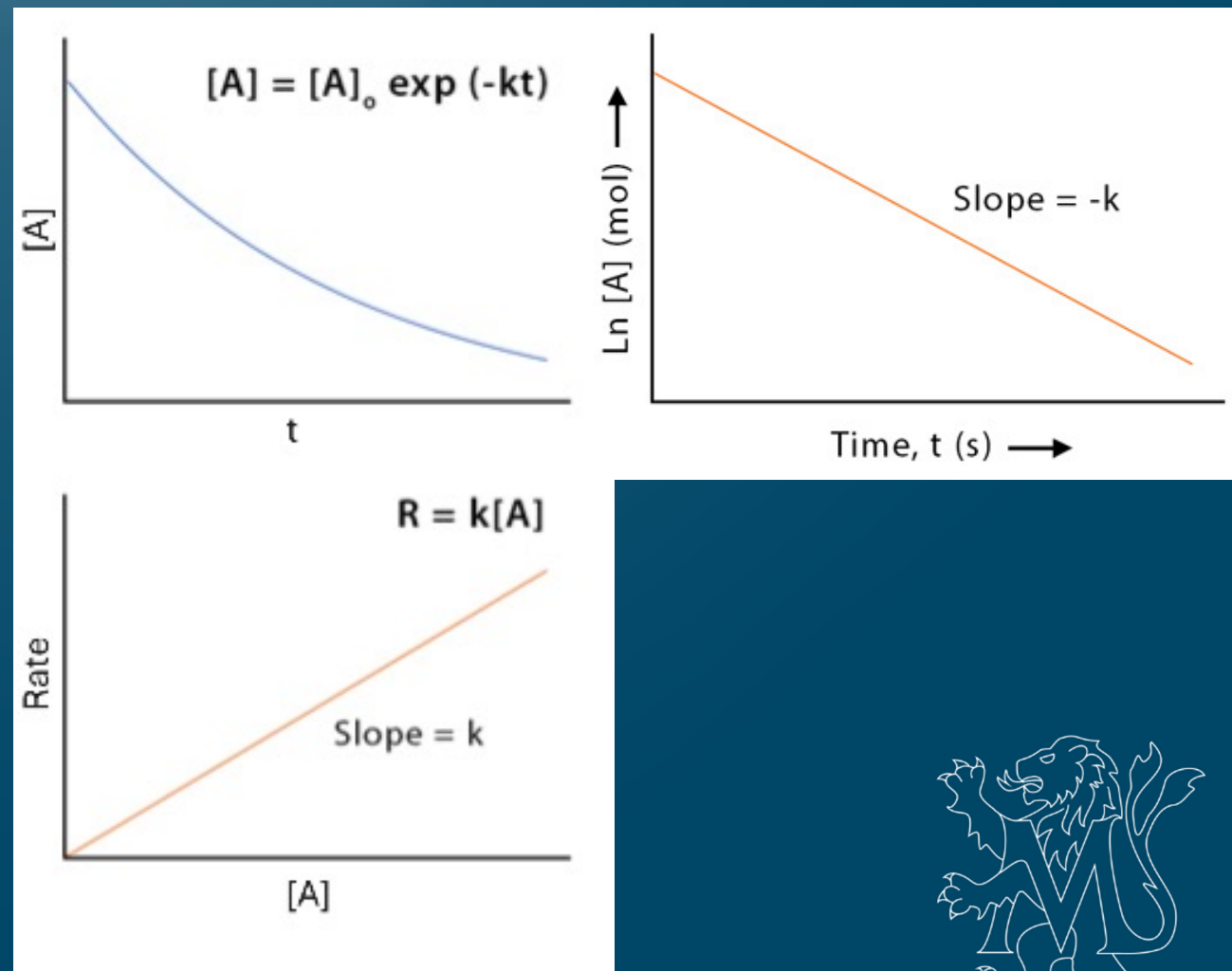
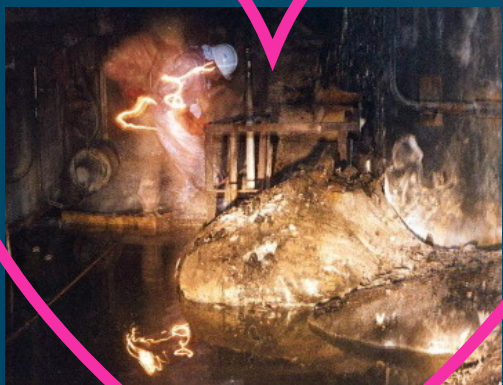
$$k \text{ (M/s)} = -m$$



1st Order



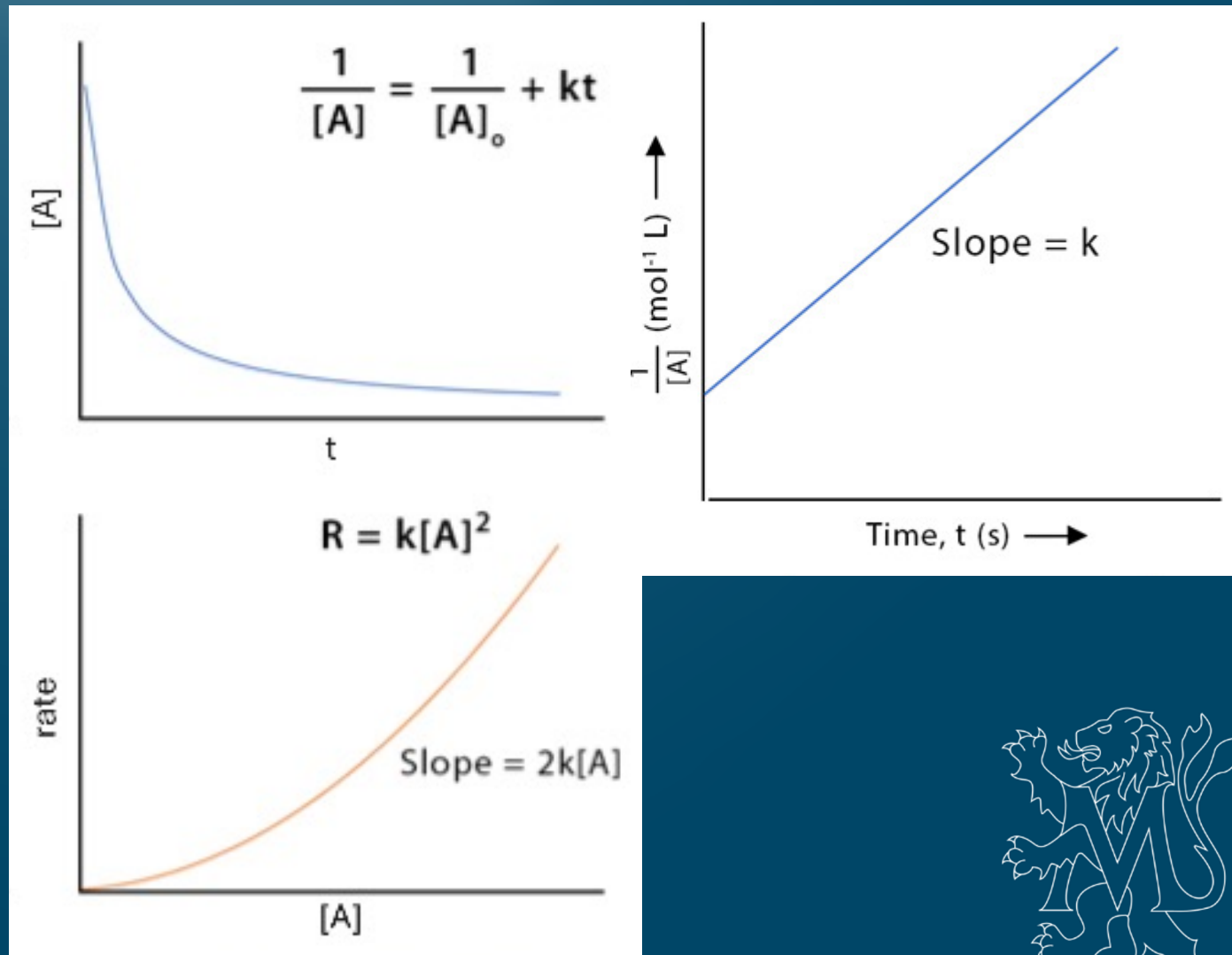
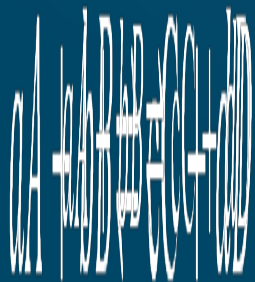
$$rate = k[A \text{ or } B] = -\frac{\Delta[A \text{ or } B]}{\Delta t} \text{ (s}^{-1}\text{)}$$



2nd Order



$$\text{rate} = k[A \text{ or } B]^2 \text{ or } k[A][B] \text{ (M}^{-1}\text{s}^{-1}\text{)}$$



Mixed Order



$$rate = \text{whatever e.g. } \frac{k_1[C][A]^2}{k_2 + k_3[A]}$$

Just estimate based on size of given terms



Question

A researcher investigates the decomposition of hydrogen peroxide in aqueous solution:



The reaction is catalyzed by iodide ions. Initial-rate measurements are collected at constant temperature, and the results are shown below.

Experiment	$[H_2O_2]$ (M)	$[I^-]$ (M)	Initial rate (M/s)
1	0.10	0.010	2.0×10^{-4}
2	0.20	0.010	4.0×10^{-4}
3	0.10	0.020	8.0×10^{-4}

The researcher proposes the rate law:

$$\text{Rate} = k[H_2O_2]^m[I^-]^n$$

Question

Based on the experimental data, which of the following is the approximate value of the rate constant, k , and its units?

- A. $20 \text{ M}^{-1} \text{ s}^{-1}$
- B. $0.20 \text{ M}^{-2} \text{ s}^{-1}$
- C. $2.0 \times 10^{-3} \text{ M}^{-1} \text{ s}^{-1}$
- D. $2.0 \times 10^{-2} \text{ M}^{-2} \text{ s}^{-1}$



General Chemistry Chapter VI

Equilibrium



6.1: Equilibrium



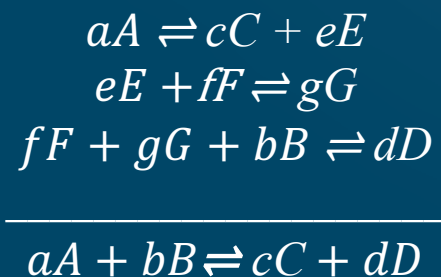
$$\Delta G = \Delta H - T\Delta S$$



Law of Mass Action



$$K_{eq} = \frac{[C]^c [D]^d}{[A]^a [B]^b} = \frac{k_f}{k_r}$$



$$K_{eq} = \frac{k_1 k_2 k_3}{k_{-1} k_{-2} k_{-3}} = K_1 * K_2 * K_3$$



Reaction Quotient



$$Q = \frac{[C]^c [D]^d}{[A]^a [B]^b}$$

$Q > K$
 *Reactants*

$Q = K$
Hooray

$Q < K$
 *Products*



Question

A 1.0 L rigid container, initially containing only gases A and B at a temperature of 298K, undergoes the following reversible reaction:



The mixture reaches equilibrium at 288K and contains 0.20 mol of A, 0.10 mol of B, and 0.40 mol of C.

i. What is the value of the equilibrium constant (K_c) for this reaction?

- A. 10 M^{-1}
- B. 1 M^{-2}
- C. 100 M^{-2}
- D. 100 M^{-1}

ii. What is the value of K_c for the reverse reaction?

- A. 10 M^{-1}
- B. 1 M^2
- C. 0.01 M^2
- D. 100 M^{-2}



Question

A researcher studies the decomposition of compound A according to the reaction:



At a particular temperature, the equilibrium constant is:

$$K_c = 4.00 \times 10^{-2}$$

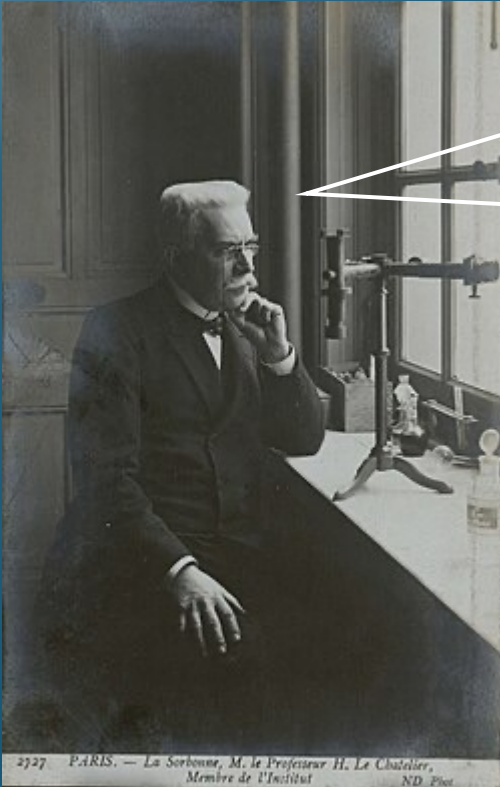
A sealed 1.00 L vessel initially contains 0.500 M of A and no B .
The reaction proceeds until equilibrium is established.

Which of the following is the closest approximation for the equilibrium concentration of B ?

- A. 0.0050 M
- B. 0.010 M
- C. 0.020 M
- D. 0.040 M



6.2: Le Chatelier's Principle



Tout système en équilibre chimique stable soumis à l'influence d'une cause extérieure qui tend à faire varier soit sa température, soit sa condensation (pression, concentration, nombre de molécules dans l'unité de volume) dans sa totalité ou seulement dans quelques-unes de ses parties, ne peut éprouver que des modifications intérieures, qui, si elles se produisaient seules, amèneraient un changement de température ou de condensation de signe contraire à celui résultant de la cause extérieure.

“When a reaction in equilibrium is disturbed, it shifts to reestablish equilibrium.”



Changes to Equilibrium

Reactant/Product quantity

$$K_{eq} = \frac{[C]^c[D]^d}{\textcircled{[A]^a[B]^b}}$$

Pressure, Volume

$$K_{eq} \neq \textcircled{\frac{[C]^c[D]^d}{[A]^a[B]^b}}$$

Temperature

$$\textcircled{K_{eq}} = \frac{[C]^c[D]^d}{[A]^a[B]^b}$$



Question

A 1.0 L rigid container, initially containing only gases A and B at a temperature of 298K, undergoes the following reversible reaction



The mixture reaches equilibrium at 288K and contains 0.20 mol of A, 0.10 mol of B, and 0.40 mol of C.

a. Suppose that a mischievous student heats this container to 308K. How will the quantity of C be affected once equilibrium is reestablished?

- A. It will increase
- B. It will decrease
- C. It will stay the same
- D. Not enough information

b. The mischievous student's angry lab partner throws the container really hard, so it dents and is now a 0.5L container. How will the quantity of C be affected once equilibrium is reestablished?

- A. It will increase
- B. It will decrease
- C. It will stay the same
- D. Not enough information



Question (Combined Chapter)

In a proposed mechanism for the decomposition of compound A , the reaction occurs in two steps:



Step 1 is a **rapid equilibrium**, while Step 2 is the **rate-determining step**. At a particular temperature, the equilibrium constant for Step 1 is:

$$K_{eq} = 4.0 \times 10^2$$

For an experiment, the equilibrium concentrations of A and B were measured to be:

$$[A] = 0.20 \text{ M}$$

$$[B] = 0.050 \text{ M}$$

The initial rate of formation of P was measured to be:

$$\text{Rate} = 2.0 \times 10^{-3} \text{ M/s}$$

Assume that Step 2 is first order with respect to I .

Question 1

What is the equilibrium concentration of intermediate I ?

- A. $2.0 \times 10^{-2} \text{ M}$
- B. $4.0 \times 10^{-1} \text{ M}$
- C. $4.0 \times 10^0 \text{ M}$
- D. $8.0 \times 10^2 \text{ M}$



Question (Combined Chapter)

In a proposed mechanism for the decomposition of compound A , the reaction occurs in two steps:



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$$\text{Rate} = 2.0 \times 10^{-3} \text{ M/s}$$

Assume that Step 2 is first order with respect to I .

Question 2

Which expression correctly represents the rate of formation of P ?

A. $\text{Rate} = K_{eq}[A][B]$

B. $\text{Rate} = k[I]$

C. $\text{Rate} = k[A][B]$

D. $\text{Rate} = K_{eq}k[A][B]$



Question (Combined Chapter)

In a proposed mechanism for the decomposition of compound A , the reaction occurs in two steps:



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$$\text{Rate} = 2.0 \times 10^{-3} \text{ M/s}$$

Assume that Step 2 is first order with respect to I .

Question 3

What is the rate constant k for Step 2?

A. $5.0 \times 10^{-4} \text{ s}^{-1}$

B. $5.0 \times 10^{-3} \text{ s}^{-1}$

C. $5.0 \times 10^{-2} \text{ s}^{-1}$

D. $5.0 \times 10^{-1} \text{ s}^{-1}$



Question (Combined Chapter)

In a proposed mechanism for the decomposition of compound A , the reaction occurs in two steps:



Step 1 is a **rapid equilibrium**, while Step 2 is the **rate-determining step**. At a particular temperature, the equilibrium constant for Step 1 is:

$$K_{eq} = 4.0 \times 10^2$$

For an experiment, the equilibrium concentrations of A and B were measured to be:

$$[A] = 0.20 \text{ M}$$

$$[B] = 0.050 \text{ M}$$

The initial rate of formation of P was measured to be:

$$\text{Rate} = 2.0 \times 10^{-3} \text{ M/s}$$

Assume that Step 2 is first order with respect to I .

Question 4

Which rate law would be predicted for the **overall reaction** based on this mechanism?

A. $\text{Rate} = k[A][B]$

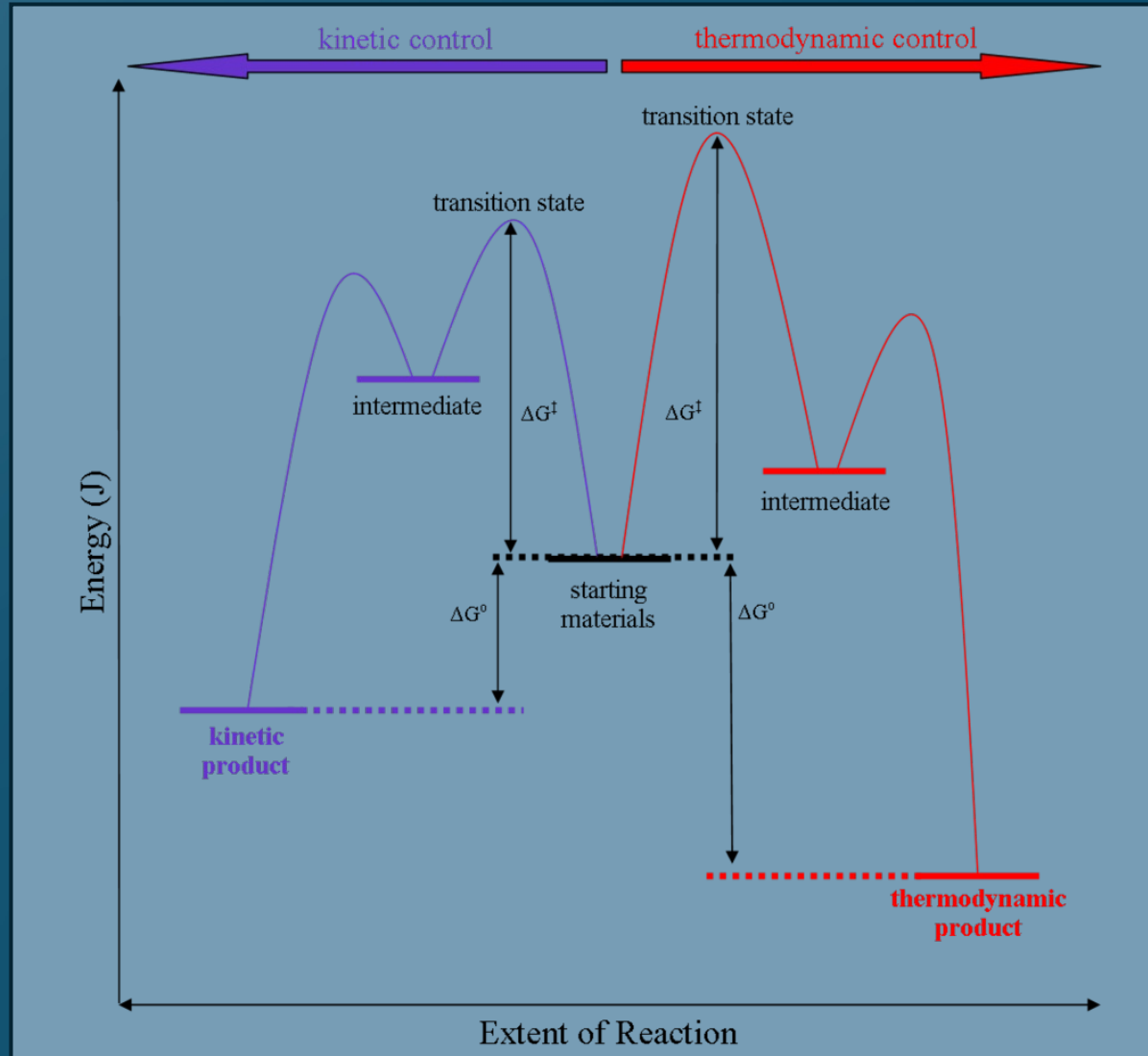
B. $\text{Rate} = k[A + B]$

C. $\text{Rate} = k[I]$

D. $\text{Rate} = k[A]^2[B]$



6.3: Kinetic and Thermodynamic Control



Question

A student is planning on taking the MCAT in May. Which course of action could be likened to a thermodynamic product?

- A. Cram study the day before and get a 495
- B. Dedicate many, many hours to studying and get a 522
- C. Force a twin to take the test
- D. Say, “I recognize I would be the most mentally stable ($\Delta G=0$) if I studied the vast majority of content, but right now the ratio of content I’ve studied compared to content I haven’t studied is lower than ideal ($Q<K$); therefore, I must push toward studying more (shift toward product)”



Combination!

MCAT Style Questions

